

Curriculum Vitae of Guolei ZHONG (仲国磊)

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Research Interest

My primary research interests lie in complex algebraic geometry and its interactions with dynamics. Particular topics include:

1. Canonical heights and Kawaguchi-Silverman conjecture
2. Equidistribution of periodic subvarieties with small dynamical degree
3. Finite morphisms between projective (especially Fano) manifolds
4. Positivity of tangent bundles of smooth projective varieties

Appointment

Mar 2026 –	Research Fellow, East China Normal University , China
Dec 2024 – Feb 2026	Senior Researcher, Institute for Basic Science , Republic of Korea
Jan 2022 – Nov 2024	Postdoctoral Research Associate, Institute for Basic Science , Republic of Korea

Education

Ph.D. in Mathematics, [National University of Singapore](#) (Supervisor: [De-Qi Zhang](#)), Singapore, 2021
B.Sc. in Mathematics, [East China Normal University](#), China, 2017

Publications & Preprints

1. (with Jun-Muk Hwang) Holomorphic symplectic geometry of elliptic surfaces, 2025, submitted.
2. (with Zelong Chen and Sheng Meng) On smooth rationally connected projective threefolds of Picard number two admitting int-amplified endomorphisms, 2025, submitted. [arXiv: 2506.14274](#)
3. (with Sheng Meng) Kawaguchi-Silverman conjecture for int-amplified endomorphism, 2024, submitted. [arXiv: 2408.00566](#)
4. (with Masataka Iwai and Shin-ichi Matsumura) Positivity of tangent sheaves of projective varieties -- the structure of MRC fibrations, Algebraic Geometry (to appear), 2025. [arXiv: 2309.09489](#)
5. (with Tien-Cuong Dinh) Periodic points for meromorphic self-maps of Fujiki varieties, Indiana University Mathematics Journal, Vol. 74 (2025), no. 6, 1561–1587.

6. (with Fei Hu) Canonical heights for abelian group actions of maximal dynamical rank, *Forum of Mathematics, Sigma*, Vol. 13 (2025), Paper No. e18.
7. (with Feng Shao) Bigness of tangent bundles and dynamical rigidity of Fano manifolds of Picard number 1 (with an appendix by Jie Liu), *Mathematische Annalen*, Vol. 391 (2025), no. 2, 1731–1752.
8. (with Jia Jia and Yongnam Lee) Smooth projective surfaces with pseudo-effective tangent bundles, *Journal of the Mathematical Society of Japan*, Vol. 77 (2025), no. 1, 75–102.
9. Characterization of Q -complex tori via endomorphisms – an addendum to “Int-amplified endomorphisms of compact Kähler spaces”, *Asian Journal of Mathematics*, Vol. 28 (2024), no. 3, 345–354.
10. (with Juanyong Wang) Strictly nef divisors on singular threefolds, *International Journal of Mathematics*, 35 (2024), no. 13, Paper No. 2450054.
11. (with Jie Liu, Wenhao Ou, Juanyong Wang and Xiaokui Yang) Algebraic fibre spaces with strictly nef relative anti-log canonical divisor, *Journal of the London Mathematical Society (2)*, Vol. 109 (2024), no. 6, Paper No. e12942, 26 pp.
12. (with Feng Shao) Boundedness of finite morphisms onto Fano manifolds with large Fano index, *Journal of Algebra*, Vol. 639 (2024), 678–707.
13. Existence of the equivariant minimal model program for compact Kähler threefolds with the action of an abelian group of maximal rank, *Mathematische Nachrichten*, Vol. 296 (2023), no. 7, 3128–3135.
14. (with Jia Jia) Amplified endomorphisms of Fano fourfolds, *manuscripta mathematica*, Vol. 172 (2023), 567–598.
15. (with Sheng Meng) Rigidity of rationally connected smooth projective varieties from dynamical viewpoints, *Mathematical Research Letters*, Vol. 30 (2023), no. 2, 589–610.
16. (with Sheng Meng and De-Qi Zhang) Non-isomorphic endomorphisms of Fano threefolds, *Mathematische Annalen*, Vol. 383 (2022), no. 3-4, 1567–1596.
17. (with Yohsuke Matsuzawa, Sheng Meng, Takahiro Shibata and De-Qi Zhang) Invariant subvarieties with small dynamical degree, *International Mathematics Research Notices*, (2022), no. 15, 11448–11483.
18. Compact Kähler threefolds with the action of an abelian group of maximal rank, *Proceedings of the American Mathematical Society*, Vol. 150 (2022), no. 1, 55–68.
19. Int-amplified endomorphisms of compact Kähler spaces, *Asian Journal of Mathematics*, Vol. 25 (2021), no. 3, 369–392.
20. Totally invariant divisors of int-amplified endomorphisms of normal projective varieties, *Journal of Geometric Analysis*, Vol. 31 (2021), no. 3, 2568–2593.

Invited talks

1. Kawaguchi-Silverman conjecture for int-amplified endomorphism, Conference on Algebraic Geometry and Arithmetic Geometry, East China Normal University, Shanghai, 5 June 2026.
2. Holomorphic symplectic geometry of elliptic surfaces, Algebraic Geometry Seminar, Nanjing University, Nanjing, 27 April 2026.
3. Kawaguchi-Silverman conjecture for int-amplified endomorphism, Algebraic Geometry Workshop 2026, Tsinghua University, Beijing, 24 April 2026.

4. Dynamical toric conjecture in dimension three, Workshop on Fano Spherical varieties – 2026, Gyeongsang National University, Jinju, February 2026.
5. Holomorphic symplectic geometry of elliptic surfaces, Workshop on Algebraic Geometry and Complex Geometry in Osaka, Osaka University (Toyonaka Campus), Osaka, 26 January 2026.
6. Holomorphic symplectic geometry of elliptic surfaces, Conference on Recent Progress in Algebraic Geometry, East China Normal University, Shanghai, 15 October 2025.
7. Canonical heights for abelian group actions of maximal dynamical rank, Topology, Geometry, and Dynamics Seminar, National University of Singapore, Singapore, 27 August 2025.
8. Surjective endomorphisms of smooth rationally connected varieties (Lectures 1-3), Summer Mathematical School "Algebra and Geometry", Suzdal, 3-5 August 2025.
9. Kawaguchi-Silverman conjecture for int-amplified endomorphisms (online), Birational Geometry Seminar, UCLA, 13 June 2025.
10. On the endomorphism conjecture and its interaction with the positivity of tangent bundles, Algebraic Geometry Seminar, East China Normal University, Shanghai, 28 March 2025.
11. Dynamical restrictions on the positivity of tangent bundles (online), Workshop on birational geometry, Steklov Institute, Moscow, 5 December 2024.
12. Dynamical restrictions on the positivity of tangent bundles, Shandong University, Jinan, 16 November 2024.
13. Projective varieties with almost nef tangent sheaves and its dynamical application, Workshop on Algebraic Geometry over complex number field or in positive characteristic, Osaka Metropolitan University (Sugimoto Campus), Osaka, 18 September 2024.
14. Periodic points for meromorphic self-maps of Fujiki varieties, 2024 Seoul Workshop on Complex Geometry and Analysis, Seoul National University, Seoul, 27 August 2024.
15. Pseudo-effectiveness of tangent bundles of smooth complete intersections and weighted hypersurfaces, Algebraic Geometry Seminar (online), School of Mathematics, Sun Yat-Sen University, Guangzhou, 23 May 2024.
16. Canonical heights for abelian group actions of maximal dynamical rank, Complex Algebraic Geometry and Related Topics, Morningside Center of Mathematics, Chinese Academy of Sciences, Beijing, 26 April 2024.
17. Structure of projective varieties with almost nef or positively curved tangent sheaves, Algebraic Geometry Seminar, June E Huh Center for Mathematical Challenges (HCMC), Korea Institute for Advanced Study (KIAS), Seoul, 9 April 2024.
18. Introduction to the positivity of tangent sheaves of projective varieties, Algebraic Geometry Seminar, June E Huh Center for Mathematical Challenges (HCMC), Korea Institute for Advanced Study (KIAS), Seoul, 8 April 2024.
19. On an endomorphism conjecture of Fano manifolds of Picard number 1, Algebraic Geometry Seminar (online), YMSC, Tsinghua University, Beijing, 12 January 2024.
20. Periodic points for meromorphic self-maps of Fujiki varieties, SCV&CG Seminar, AMSS, Chinese Academy of Sciences, Beijing, 11 January 2024.
21. Canonical heights for abelian group actions of maximal dynamical rank, Algebraic Geometry Seminar, Shanghai Jiao Tong University, Shanghai, 4 January 2024.
22. Dynamical rigidity of Fano manifolds of Picard number 1 with big tangent bundle, Algebraic Geometry in East Asia, Korea Institute for Advanced Study, Seoul, 9 November 2023.

23. Positivity of tangent sheaf of projective varieties, Algebraic Geometry Seminar, Center for Complex Geometry, Institute for Basic Science, Daejeon, 10 October 2023.
24. Structure of projective varieties with certain positive tangent sheaves, Workshop on Complex Dynamics and Geometry, Institute for Mathematical Sciences (IMS), National University of Singapore, Singapore, 31 August 2023.
25. Compact Kähler threefolds with the action of an abelian group of maximal rank, Korea Several Complex Variables Workshop, Center for Complex Geometry, Institute for Basic Science, Daejeon, 16 August 2023.
26. Invariant subvarieties with small dynamical degree, 2023 KMS Spring Meeting, Special Session, Daejeon Convention Center, Daejeon, 28 April 2023.
27. Smooth projective surfaces with pseudo-effective tangent bundles, Seminars on Algebraic Surfaces and Related Topics, Center for Complex Geometry, Institute for Basic Science, Daejeon, 27 February 2023.
28. Finite morphisms onto Fano manifolds with large Fano index, Informal Seminar, Center for Complex Geometry, Institute for Basic Science, Daejeon, 21 December 2022.
29. Finite morphisms onto Fano manifolds with large Fano index, Algebraic Geometry Seminar (online), East China Normal University, Shanghai, 15 December 2022.
30. Dynamical characterization of projective toric Varieties, Complex Geometry Seminar, Center for Complex Geometry, Institute for Basic Science, Daejeon, 1 September 2022.
31. Compact Kähler threefolds with the action of an abelian group of maximal dynamical rank, Geometry Seminar, Korea Institute for Advanced Study, Seoul, 19 July 2022.
32. Compact Kähler threefolds with the action of an abelian group of maximal dynamical rank, Zoom Algebraic Geometry Seminar (online), 15 March 2022.
33. Strictly nef divisors on singular varieties, Complex Geometry Seminar, Center for Complex Geometry, Institute for Basic Science, Daejeon, 3 March 2022.
34. Strictly nef divisors on singular varieties, II, Geometry Seminar (online), Korea Institute for Advanced Study, Seoul, 25 January 2022.
35. Strictly nef divisors on singular varieties, I, Geometry Seminar (online), Korea Institute for Advanced Study, Seoul, 24 January 2022.
36. Characterization of projective toric varieties from dynamical viewpoints, Weekly seminar on Explicit Birational Geometry (online), SCMS Fudan University, Shanghai, 5 January 2022.
37. Projective varieties with (relative) anti-log canonical divisors being strictly nef, Topology and Geometry Seminar (online), National University of Singapore, Singapore, 3 November 2021.
38. Algebraic dynamics on compact normal varieties, Ph.D. Oral Defense, National University of Singapore, Singapore, 6 October 2021.
39. Fano threefolds and fourfolds admitting non-isomorphic endomorphisms, Seminar of Algebraic Geometry in East Asia (online), 5 March 2021.
40. Rigidity of rationally connected smooth projective varieties from dynamical viewpoints, Seminar on Algebraic Dynamics (online), Korea Institute for Advanced Study, Seoul, 1 July 2020.
41. Int-Amplified endomorphisms of compact Kähler spaces, Topology and Geometry Seminar, National University of Singapore, Singapore, 29 January 2020.

42. Totally invariant divisors of amplified endomorphisms of normal projective varieties, Algebraic Geometry Seminar, East China Normal University, Shanghai, 25 June 2019.
43. Totally invariant divisors of amplified endomorphisms of normal projective varieties, Topology and Geometry Seminar, National University of Singapore, Singapore, 15 May 2019.

Teaching Experiences

1. Fall 2021, GEH1036/GEK1505, Living with Mathematics (Tutor), NUS, Singapore.
2. Spring 2020, MA1513, Linear Algebra with Differential Equations (Tutor), NUS, Singapore.
3. Fall 2019, MA1513, Linear Algebra with Differential Equations (Tutor), NUS, Singapore.
4. Spring 2019, MA1513, Linear Algebra with Differential Equations (Tutor), NUS, Singapore.
5. Spring 2019, MA1512, Differential Equations for Engineering (Tutor), NUS, Singapore.
6. Fall 2018, GEH1036/GEK1505, Living with Mathematics (Tutor), NUS, Singapore.

Academic Services

1. Reviewer for Mathematical Reviews (since October 2022)
2. Reviewer for zbMATH (since May 2023)
3. Referee for Adv. Math., ANT, BKMS, BLMS, IJM, IMRN, J. Alg., JKMS, JLMS, Math. Ann., Math. Nach., PRIMS, Sci. China Math., Taiwanese J. Math., Trans. Groups

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